

Project Report

Data Mining Group 11

Can Synthetic Data Augmentation Unlock Superior Classification Performance on Small Datasets, or Does It Introduce More Noise Than Signal?

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The relevant code for this paper can be found [here](#).

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1 Problem Description

Obesity is a major global health concern associated with chronic diseases such as cardiovascular conditions and diabetes. As obesity rates rise due to urbanization, poor diet, and sedentary lifestyles, early risk prediction has become essential. Machine learning can support preventive care by identifying individuals at risk.

1.1 Application Area and Goals

This project explores the use of data mining techniques for obesity risk classification. The target variable `NOBeyesdad` represents six obesity levels, framing our task as a multi-class classification problem.

A key challenge is the limited size and imbalance of real-world health data. Synthetic data augmentation is a promising solution, though its impact remains debated, offering potential improvements in class balance and generalization, but also risks of introducing noise.

The goal of this paper is to evaluate the impact of augmenting real-life data with synthetically generated data. We aim to identify which models benefit, and whether this practice improves or harms performance and generalization.

Nomenclature In the following Paper, nomenclature is defined as

RD real dataset

PD pseudoreal dataset

SD synthetic dataset

The specifics of the datasets will be clarified in section [Data Profile](#). Additionally, when using a concatenation of multiple of the aforementioned datasets, the shorthands will be merged (e.g., `RD || PD` becomes `RPD`)¹.

1.2 Data Profile

Dataset Definitions The datasets utilized for this project were initially gathered via an online survey. To address the class imbalance in `RD`, SMOTE² was applied, resulting in the dataset `PD` ([Palechor and Manotas 2019](#)). Additionally, `SD` was generated using a deep learning model (see [Kaggle Playground Series S4E2](#)). The definitions of the datasets employed in this study are as follows:

- `RD`: 477 datapoints of human-generated survey data
- `PD`: 1,634 SMOTE-generated datapoints generated from `RD`
- `SD`: 20,759 datapoints generated from `RPD` $\Leftrightarrow$ `RD || PD` by a deep learning model

¹The operation `A || B` is defined as the concatenation of all items in set `A` with the items in set `B`

²Synthetic Minority Oversampling Technique

All 3 datasets share the same structure, consisting of 16 input attributes and one multi-class target variable `NObeyesdad`, but differ in their origin and purpose. This allows us to compare the datasets and draw a concise conclusion.

Initially, we thought that there were only two datasets, because the data was packaged in 2 instances, one instance with 2,111 samples (RPD) labeled as real and one with 20,759 samples (SD) labeled as synthetic, subsequently we discovered that RPD actually consisted of 2 different parts RD and PD. Hence, we decided to perform 2 experiments:

Experiment Training sets (D_{train})		Test set (D_{test})
Initial	RPD, RPSD	TEST_RPD = 20% of RD
Extra	RD, RPD, RPSD	TEST_RD = 20% of RPD

For each experiment, the following holds: $\forall D_{\text{train}} \in \text{Training sets}, \forall D_{\text{test}} \in \text{Test set}: D_{\text{train}} \cap D_{\text{test}} = \emptyset$

Table 1: Experiment definition

From this point onward, all graphs and results presented will follow the naming convention of the Initial and Extra experiments presented in [Table 1](#). This way it is clear which test set we are talking about. The idea of having a consistent dataset over different training set is the only way to see the impact of augmenting data by synthetic means.

1.2.1 Target Variable Distribution

The target variable, `NObeyesdad`, is a multi-class categorical label representing obesity risk levels based on BMI, as defined by ([Palechor and Manotas 2019](#)). The categories are defined as:

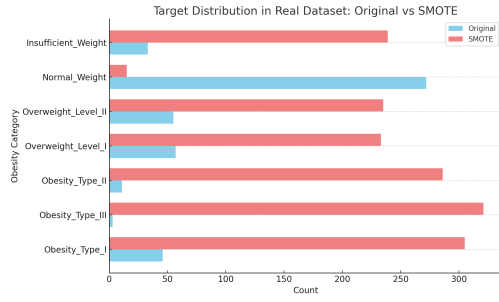
$$\text{NObeyesdad} = \begin{cases} \text{Underweight} & \text{if BMI} < 18.5 \\ \text{Normal_Weight} & \text{if } 18.5 \leq \text{BMI} < 25 \\ \text{Overweight} & \text{if } 25 \leq \text{BMI} < 30 \\ \text{Obesity_Type_I} & \text{if } 30 \leq \text{BMI} < 35 \\ \text{Obesity_Type_II} & \text{if } 35 \leq \text{BMI} < 40 \\ \text{Obesity_Type_III} & \text{if BMI} \geq 40 \end{cases}$$

To analyze class imbalance and the effects of synthetic data generation, we visualized the distribution of the target variable for RD, PD and SD in [Figure 1](#).

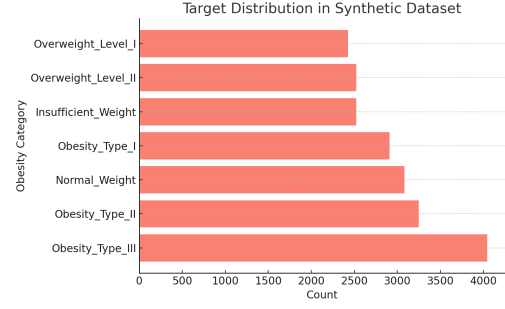
1.2.2 Attributes and Types

The datasets contain 16 features used to predict the categorical `NObeyesdad` target variable. The attributes are as follows:

Important Attributes `Weight` and `Height` are fundamentally linked to BMI and thus obesity categories. `Age`, dietary habits (`FAVC`, `FCVC`, `CAEC`), and physical activity (`FAF`) are also commonly considered significant predictors in obesity studies ([Khater et al. 2023](#); [Ferrerias et al. 2023](#); [Safaei et al. 2021](#)). `Height` and `Weight` are potential sources of multicollinearity and candidates for feature selection.



(a) Target variable distribution in RD, PD



(b) Target variable distribution in SD

Figure 1: Target variable distributions in the different datasets

Feature	Data type	Encoding/Scaling	Description
Age	Numeric	Standard scaling with rounding to int	Age
Height	Numeric	Standard scaling with rounding to 2 decimals	Height
Weight	Numeric	Dropped (false predictor with Height/BMI)	Weight
family_history_with_overweight	Binary	Map values to (0,1)	Family history with overweight
FAVC	Categorical	Map values to (0,1)	Frequent consumption of high-caloric food
FCVC	Categorical	Ordinal, with rounding to integer	Frequency of vegetable consumption
NCP	Numeric	Ordinal, with rounding to integer	Number of main meals per day
CAEC	Categorical	Map values to (0,1,2,3)	Consumption of food between meals
CH2O	Categorical	Ordinal, with rounding to integer	Daily water consumption
CALC	Categorical	Map values to (0,1,2,3)	Alcohol consumption
SMOKE	Binary	Dropped (mostly False)	Smoking habit
SCC	Binary	Map values to (0,1)	Calories monitoring
FAF	Categorical	Ordinal, with rounding to integer	Physical activity per week
TUE	Categorical	Ordinal, with rounding to integer	Frequency of using tech devices
MTRANS	Categorical	One-Hot	Mode of transportation
NObeyesdad (Target variable)	Categorical	Map values to (0,1,2,3,4,5,6)	Obesity classification

Table 2: Features, data types, encodings, and descriptions.

1.2.3 Exploratory Data Analysis

To better understand the relationship between key biometric variables and obesity levels, we analyzed the distribution of selected important features like **Weight**, **Height**, and **Age** across the different target classes in all datasets.

Weight Distribution The weight distribution shows a clear upward trend across increasing obesity levels in all three datasets. Median values and interquartile ranges rise consistently from **Insufficient_Weight** to **Obesity_Type_III**, confirming weight as a strong predictor of obesity classification. **SD** exhibits a slightly wider spread and more variability in extreme categories, likely due to the generative model’s approximation of real-world variation. Outliers are present but aligned with expected distributions.

Height Distribution **Height** does not show a strong association with obesity level. The distributions across all categories remain relatively stable with minor fluctuations. Although there are slight median increases in some higher obesity classes, these are

not consistent or significant. Thus, height alone is not a strong predictor of obesity classification but may contribute in combination with other features like weight or age.

Feature	RD	PD	SD	RPD	RPSD
Age	0.134	0.324	0.350	0.296	0.345
CAEC	0.080	0.457	0.332	0.352	0.334
CALC	0.000	0.304	0.310	0.225	0.250
CH2O	0.082	0.188	0.313	0.235	0.304
FAF	0.061	0.190	0.261	0.212	0.256
FAVC	0.044	0.356	0.273	0.328	0.278
FCVC	0.061	0.307	0.325	0.293	0.320
Gender	0.082	0.630	0.619	0.556	0.613
Height	0.051	0.220	0.268	0.204	0.261
MTRANS	0.058	0.263	0.167	0.179	0.168
NCP	0.120	0.264	0.220	0.245	0.222
SCC	0.116	0.251	0.222	0.235	0.222
SMOKE	0.158	0.181	0.101	0.111	0.101
TUE	0.089	0.190	0.250	0.217	0.245
Weight	0.530	0.622	0.648	0.575	0.595
family_history_with_overweight	0.215	0.607	0.556	0.540	0.555

Table 3: Correlation of features with the target variable NObeyesdad in the different datasets.

Correlation Analysis After carefully inspecting each dataset and the target variable, we observed significant differences in the distributions of the target and feature values. Presumably, those inconsistencies may have been introduced by the SMOTE data generation process. We hypothesize that the chosen value of $k_neighbors$ ³ was comparatively high, causing the majority values to overpower the minority values. An example of this effect is depicted in Figure 2, here we observe a noticeable shift in the distribution of the CAEC feature with respect to the target. The value “*Sometimes*” (beige) does overpower the rest, especially if we compare its distribution shift in “*Overweight_Level_I*”, “*Obesity_Type_I*”, “*Obesity_Type_II*”. In those cases the change is easily observable. This leads to the documented values in Table 3.

2 Preprocessing

2.1 Similarities removal

Since the SD was generated before the train/test split, it leads to a data leakage issue for our synthetic records being derived from records in test sets. To reduce this data leakage, the Gower distance metric has been used. It works on raw data and finds similarities to

³This parameter decides how many neighboring datapoints are taken into account when generating SMOTE data (see [the documentation of imbalanced-learn](#))

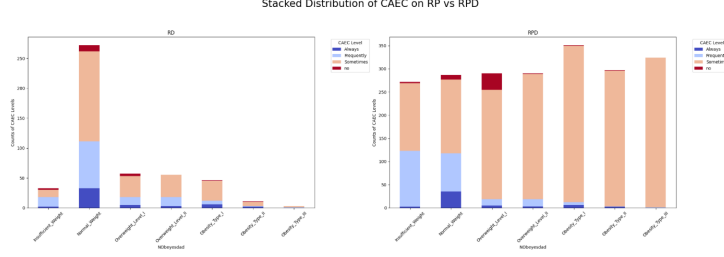


Figure 2: Stacked Feature Value Distribution for Feature CAEC

some threshold. We based our threshold selection on the number of similarities found to be:

- At least equal to the size of the test sets (each record from the test set must appear at least once in the SD)
- Not too large, since our datasets are already small

The those found similarities were removed from the SD based on two test sets for Initial and Extra experiments correspondingly.

2.2 Feature encoding

Prior to modeling, a thorough preprocessing pipeline was applied to ensure the integrity, consistency, and usability of the data in all datasets.

The dataset was found to be complete with no missing values. Only one record was removed as noise, since it's value for **CALC** feature appeared only once in all datasets. In addition, 9 unique values were found to have duplicates, totaling 34 entries. Since our datasets are quite small we decided to keep those values, but include them only in the train sets only.

Feature Transformation For transformation, a pipeline was created using a **Column-Transformer** to process the feature groups appropriately. The PD dataset contained wrong data types for some features (e.g. 21.444 for Age), those features were rounded during preprocessing. Corresponding preprocessing methods are mentioned in [Table 2](#).

The results of the preprocessing step were saved to the corresponding CSV files.

3 Data Mining Approaches

Initial Classifier Evaluation To systematically explore a range of classification strategies, we employed nine different machine learning models to explore a range of classification strategies, encompassing both symbolic and non-symbolic approaches. Decision Trees represent the symbolic category. Random Forest and Gradient Boosting (including both XGBoost and LightGBM) classifiers extend this concept through ensemble learning. For simple non-symbolic models, we include k -Nearest Neighbors (k -NN) and Naive

Bayes. Both provide useful baselines and allow for straightforward comparison with more sophisticated techniques. To further assess more complex non-symbolic methods, we employ Support Vector Machines (SVM) and a Multi-Layer Perceptron (MLP) classifier. This diverse model selection enables a comprehensive evaluation of classification performance across varying algorithmic principles and complexity levels. To benchmark performance, a baseline model that always predicts the majority class in the training data was included, which provides a reference point for evaluating whether the trained models outperform naive guessing⁴.

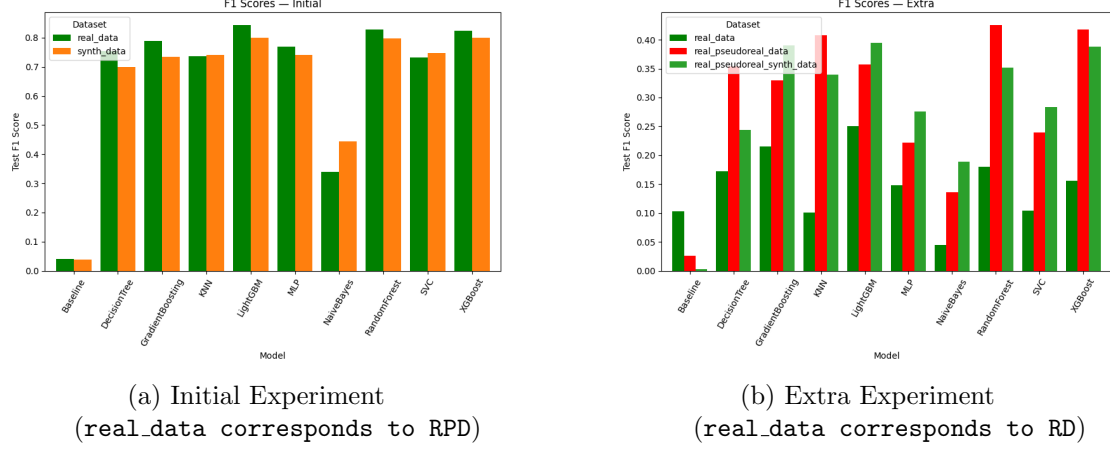


Figure 3: F1-Scores of untuned Models

All untuned models outperformed the baseline with the notable exception of Naive Bayes which was trained on RD in the Extra Experiment. This underperformance can be attributed to fundamental limitations of the Naive Bayes framework. Specifically, the model suffers from poor prior estimation, as several classes in the dataset are heavily underrepresented, leading the model to systematically underpredict these categories. Additionally, the likelihood estimates become unreliable due to the small number of samples per class, resulting in high variance and unstable predictions. These limitations highlight the sensitivity of Naive Bayes to class imbalance and low-data regimes, particularly when deployed on different or extrapolated data in terms of distribution.

Hyperparameter Tuning To optimize model performance, the tuning process was carried out using 25 iterations of `RandomizedSearchCV`, due to its speed and scalability with larger parameter spaces, particularly for models like Decision Tree, ensemble methods, and neural networks. Leave-One-Out Cross-Validation for training on RD and stratified 5-Fold Cross-Validation for training on all of the other training sets were applied consistently across all classifiers to ensure a balanced representation of classes, addressing the aforementioned issue of class imbalance and minimizing the risk of data leakage.

⁴Naive guessing here is defined as the concept of just remembering which value was seen the most often and always guessing it.

Scoring was based on macro-averaged F1-score to account for class imbalance. The best model from each search was then retrained on the full training set and evaluated on the held-out test set. A custom baseline model, always predicting the majority class without tuning, served as a naive reference point to highlight the challenges posed by class imbalance.

Model	Tuned Hyperparameters (Tried Values)
RandomForest	n_estimators $\in \{100, 200, 500\}$, max_depth $\in \{\text{None}, 3, 5, 7\}$, min_samples_split $\in \{2, 5\}$, min_samples_leaf $\in \{1, 2\}$
GradientBoosting	n_estimators $\in \{100, 200, 500\}$, learning_rate $\in \{0.01, 0.1, 0.2\}$, max_depth $\in \{3, 5, 7\}$
XGBoost	n_estimators $\in \{100, 200, 500\}$, learning_rate $\in \{0.1, 0.2\}$, max_depth $\in \{3, 5, 7\}$
kNN	n_neighbors $\in \{3, 5, 10, 20\}$, weights $\in \{\text{uniform}, \text{distance}\}$
LGBM	num_leaves $\in \{30, 50, 100, 150\}$, reg_alpha $\in \{0.1, 0.5\}$, lambda_l1 $\in \{0, 1\}$
Decision Tree	max_depth $\in \{\text{None}, 5, 10\}$, min_samples_split $\in \{2, 5\}$, criterion $\in \{\text{gini}, \text{entropy}\}$
MLPClassifier	hidden_layer_sizes $\in \{(50,), (100,), (50, 50), (100, 50)\}$, activation $\in \{\text{relu}, \text{tanh}, \text{logistic}\}$, alpha $\in \{0.0001, 0.001, 0.01\}$, learning_rate $\in \{\text{constant}, \text{adaptive}\}$
Baseline	—

Table 4: Tuned hyperparameters and corresponding value ranges for each model

Tuned Model Evaluation To evaluate the effectiveness of synthetic data in enhancing classification performance, we conducted two experimental analyses that employ distinct training and testing configurations. For the definitions of the two experiments as well as the test and train sets please refer to [Table 1](#).

4 Evaluation

4.1 Evaluation Setup

The evaluation setup for this project is designed to assess model performance across multiple data configurations, while avoiding data leakage and maintaining fairness in comparisons. The dataset is split and utilized through two primary experiments: the Initial experiment and the Extra experiment.

The Initial setup uses RPD and RPSD as training datasets, with the test set TEST_RPD. This setup explores how the inclusion of synthetic data influences model performance when tested on a dataset composed of real-world data and by SMOTE generated data. We adopted this approach, since a lot of literature that cite ([Palechor and Manotas 2019](#)) do take the dataset and treat it as completely real. The Extra setup extends this analysis by varying the training datasets more finely across RD, RPD, and RPSD, while using TEST_RD as the test set. This allows a nuanced understanding of how each augmentation layer contributes to model effectiveness.

Stratification is applied to all train-test splits to maintain the original class distribution

of the target variable. This is particularly crucial in a multi-class setting with imbalanced data, ensuring a fair and representative evaluation.

Evaluation Metric F1 score is chosen as the primary evaluation metric for model development and optimization as it is a balanced measure that accounts for both false positives and false negatives.⁵

4.2 Tuned Model Evaluation

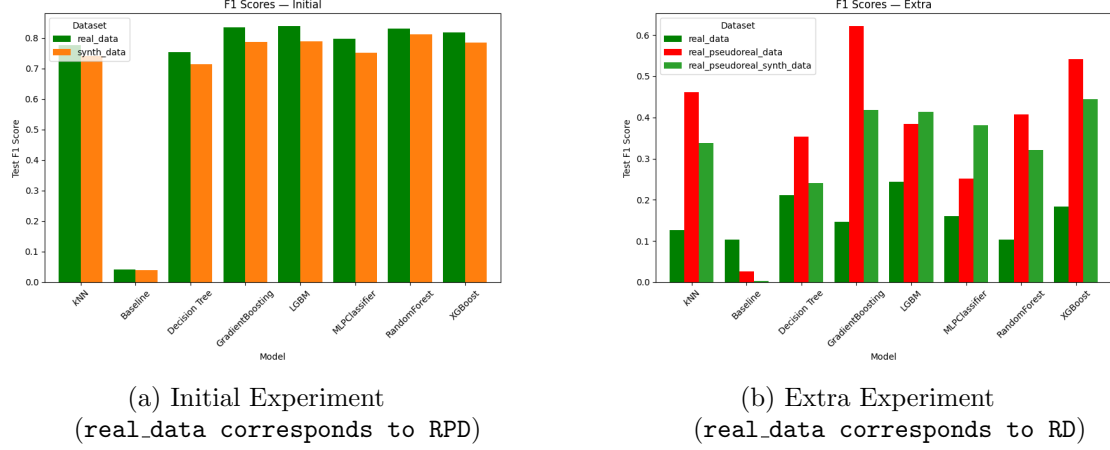


Figure 4: F1-Scores of tuned Models

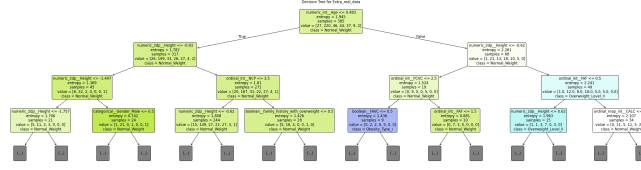
Initial Experiment In the Initial experiment, all models performed relatively similar and solid. The performance was consistently better for the RPD data and decreased for the augmented dataset RPSD. We assume that the augmentation added more noise, which lead to poorer generalization.

Extra Experiment As the result of our Extra experiment we saw that the general performance decreased significantly since everything was evaluated on a smaller dataset with a way different target class distribution. However, models trained on RD had the worst performance in comparison to the models trained on RPD and RPSD due to small training dataset size and high imbalance of target classes. In the case where we augmented the training data with PD, the performance significantly increased as expected due to a much better balance of target classes and increase in overall samples. When augmenting the data further with SD performance decreased again since the deep learning algorithm, which was used to generate the additional synthetic data, just added noise. Apart from

⁵Macro-averaged F1-score was chosen to evaluate models in a class-agnostic manner, ensuring that minority classes were not overshadowed by dominant ones, which is a common pitfall when using accuracy or support-weighted metrics in imbalanced settings.

these observations, it is also visible that certain models (c.f., LGBM, MLP) perform better when augmenting with more synthetic data which can probably be attributed to their better resistance to noise (e.g., the Multi-Layer Perceptron learning to filter noise via its non-linear activation functions).

Error Analysis During analysis of confusion matrices and feature importance for different model-dataset combinations, we saw that the feature importance differs not only across different models but also across the different datasets used. However, since we used multiple models and have 3 datasets in Extra experiment, it limits our ability to cover all of them in this report. However it is safe to say that most models include **Height** and **Age** in their most important features for decision making.



Truncated to the first $n = 3$ layers

Figure 5: Trained and Tuned Decision Tree on RD

When analyzing the confusion matrices (Figure 6) for the decision tree, it is clearly observable that first of all the distribution of the test data is very **Normal.Weight**-heavy, which is attributed to the generation of test data only from RD. In the first matrix, the model which is trained on the remainder of data in RD, a very small training set, consequently acts mostly like a model that mostly only saw examples from class **Normal.Weight** and mostly classifies anything as **Normal.Weight**. The second matrix shows a way more complex model that actually predicts more diversely and learned more than just the majority class. The final matrix shows the confusion matrix for a Decision Tree that was trained on even more noisy data than the second one and consequently performs a bit worse with more wrong predictions made.

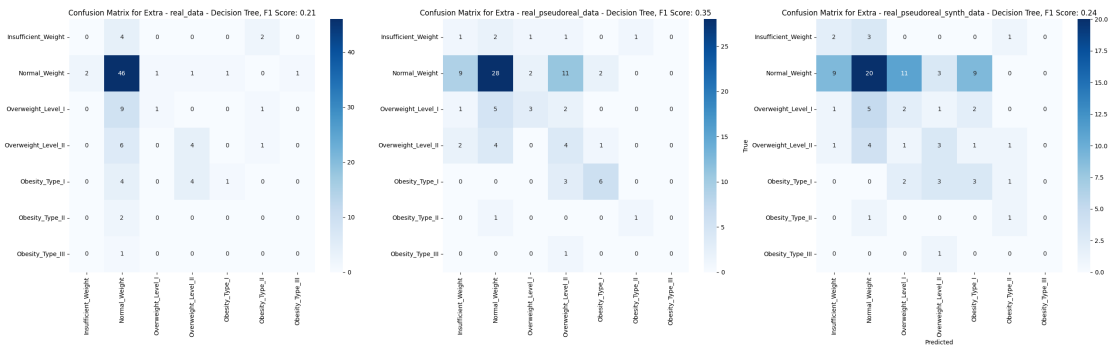


Figure 6: Decision Tree Confusion Matrix (Extra Experiment)

5 Conclusion

This study examined whether synthetic data augmentation can enhance classification performance on small datasets in the context of obesity risk prediction. Our experiments, conducted across multiple machine learning models and data configurations, reveal several important findings.

Synthetic data augmentation produces mixed results that are highly model-dependent. For ensemble methods like XGBoost and LightGBM, augmentation with synthetic data (RPSD) generally improved performance compared to training on real data alone (RD), with F1-score increases of 5-10% in the Extra experiment. This suggests that these models can effectively leverage the additional patterns provided by synthetic samples while remaining robust to potential noise.

Conversely, tree-based models like Decision Trees and Random Forest showed worse performance when trained on synthetic data, often defaulting to majority class predictions and failing to capture minority class distinctions. This indicates that simpler symbolic models may be more susceptible to the noise introduced by synthetic generation processes. The k -NN algorithm demonstrated reasonable consistency across datasets but showed sensitivity to distribution shifts between real and synthetic data.

Regarding our central research question, synthetic data can indeed unlock superior classification performance, but only when: (1) paired with appropriate model architectures that can handle potential noise, and (2) when the generation process preserves meaningful patterns from the original data. The SMOTE-generated pseudo-real data (PD) proved more beneficial than purely synthetic data (SD) for most models, suggesting that simpler augmentation techniques may sometimes outperform more complex generative approaches.

Our work has limitations, particularly in the small size of the original dataset (RD) and potential data leakage concerns despite our precautions. Future research should explore more sophisticated synthetic data filtering techniques, investigate hybrid approaches combining multiple generation methods, and verify findings across additional domains beyond obesity classification.

In practical terms, practitioners working with small healthcare datasets may benefit from augmentation strategies, but should carefully validate model performance on purely real data and consider model-specific responses to synthetic inputs. The significant performance variation observed across model types emphasizes the importance of comprehensive model evaluation when working with augmented datasets.

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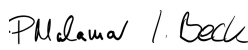
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Ehrenwörtliche Erklärung

Ich versichere, dass ich die beiliegende Projektarbeit ohne Hilfe Dritter und ohne Benutzung anderer als der an...gegebenen Quellen und in der untenstehenden Tabelle angegebenen Hilfsmittel angefertigt und die den benutzten Quellen wörtlich oder inhaltlich entnommenen Stellen als solche kenntlich gemacht habe. Diese Arbeit hat in gleicher oder ähnlicher Form noch keiner Prüfungsbehörde vorgelegen. Ich bin mir bewusst, dass eine falsche Erklärung rechtliche Folgen haben wird.

Declaration of Used AI Tools			
Tool	Purpose	Where?	Useful?
ChatGPT	Rephrasing, More Concise Wording	Throughout	+
DeepL	Synonym Search, Paraphrasing	Throughout	++
Claude 3.7 Sonnet	Downscaling Tables	Table 3	-



Signature

Mannheim, May 19, 2025